MURALIDHAR REDDY ANUMANDLA

Data Engineer | Azure & Databricks Data Pipelines

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SUMMARY

Cut a 140M-row Spark job from a 20-hour crash to 2-4 hours. Data Engineer on regulated healthcare data at UnitedHealth Group: own Azure Data Factory and Databricks ETL/ELT pipelines, monitor 25+ in production, and fix the data defects analysts hit. M.S. Computer Science, VCU.

SKILLS

Languages & Querying: SQL / T-SQL (window functions, tuning, indexing), Python, PySpark, Spark SQL, Pandas

Warehousing & Modeling: Dimensional modeling (star schema, fact/dimension), ETL/ELT, medallion architecture, incremental loads, upsert/merge, data quality validation, partitioning, REST API ingestion

Platforms & Cloud: Azure Databricks, Delta Lake, Unity Catalog, Azure Data Factory, Databricks Workflows, dbt, Apache Airflow, Kafka, Spark Structured Streaming, Azure SQL Database, Blob Storage, Key Vault

Engineering Practices: Git/GitHub (PR review), GitHub Actions CI, CI/CD, Agile/Scrum, Rally, GitHub Copilot, Claude Code

AI & Reporting: LLM tool-calling (Ollama), embeddings / vector store, Power BI (DAX), Excel (pivots, charts)

EXPERIENCE

UNITEDHEALTH GROUP (Client), Remote — Jun 2025 – Present

Data Engineer (Feb 2026–Present) | Data Engineer Intern (Jun–Dec 2025)

- Cut a failing Spark job on ~140M rows from a 20-hour OOM crash to a 2-4 hour run (~5-10x faster) with window-function dedup, early filtering, and Parquet staging.
- Own ETL/ELT pipelines end to end on Azure Data Factory and Databricks; monitor 25+ in production for SLA misses.
- Fixed duplicate member-coverage records in an 8-table, Kafka-fed streaming pipeline (faulty derived column and merge key), restoring clean analyst tables; flagged the source-side cause to the analytics head.
- Built a Kafka-fed Spark Structured Streaming pipeline into Delta tables, decrypting PII via Key Vault and publishing unsubscribe flags that control who the email platform can contact.
- Contributed to migrating ~3.6M legacy records from one all-or-none consent flag to a category-level model, helping close a compliance gap that risked wrongly emailing or suppressing users.
- Fixed a stalled staging file-ingestion pipeline owned by another developer: traced it to a cluster upgrade that broke pipeline code, cleared the follow-on defects it exposed, and explained the root cause to business users.
- Validated a deduplication feature across 20 demographics tables (team of four, 6 tables personally) and 8 tables post-migration using reusable, parameterized SQL checks.

PERSONAL PROJECTS

NYC MOBILITY INCREMENTAL PIPELINE

PySpark, Delta Lake MERGE, dbt (DuckDB), Airflow, GitHub Actions; 48.7M real NYC TLC trips plus weather, run locally

- Built an incremental bronze/silver/gold pipeline over 48.7M real NYC taxi trips with an idempotent Delta MERGE per month; quarantined 978,679 bad rows (2.0%) with reasons and reconciled bronze = silver + quarantine.
- Late-arriving data test: replay inserted 19,380 and updated 4,938 rows; 0 differing rows vs a clean load, 0 duplicates.
- dbt gold layer (37 of 37 tests pass) plus an Airflow DAG; GitHub Actions CI runs lint, pytest, the pipeline and dbt build.

CMS CLAIMS LAKEHOUSE

PySpark, Delta Lake, dbt (DuckDB), GitHub Actions; real CMS Medicare public data (1.56M rows), run locally

- Built a bronze/silver/gold lakehouse on two real CMS Medicare files (1.56M rows) with quarantine and row reconciliation.
- Silver written by Delta MERGE on the business key: a re-run left counts unchanged, 0 duplicate keys; CI repeats it.
- dbt star schemas with 32 data tests (42 of 42 steps pass); found a 2.1x state payment gap for hip/knee DRG 470.

EDUCATION

VIRGINIA COMMONWEALTH UNIVERSITY, Richmond, VA: M.S. Computer Science, GPA 3.90/4.0 — 2024 – 2025

GRIET, Hyderabad, India: B.Tech Computer Science & Business Systems, GPA 8.31/10 — 2020 – 2024